RAJALAKSHMI ENGINEERING COLLEGE

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R2

Started on	Thursday, 8 October 2026, 10:58 AM
State	Finished
Completed on	Thursday, 8 October 2026, 10:58 AM
Time taken	45 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Find Duplicate in Array.

Given a read only array of n integers between 1 and n, find one number that repeats.

Input Format:

First Line - Number of elements

n Lines - n Elements

Output Format:

Element x - That is repeated

For example:

Input	Result
5 1 1 2 3 4	1

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

int findDuplicate(const int* A, int n) {
    if (n <= 1) return -1;

    int slow = A[0];
    int fast = A[0];

    do {
        slow = A[slow];
        fast = A[A[fast]];
    } while (slow != fast);

    slow = A[0];
    while (slow != fast) {
        slow = A[slow];
        fast = A[fast];
    }

    return slow;
}

int main() {
    int n;
    if (scanf("%d", &n) != 1) return 0;

    int A[n];
    for (int i = 0; i < n; i++) {
        scanf("%d", &A[i]);
    }

    printf("%d\n", findDuplicate(A, n));
    return 0;
}

```

	Input	Expected	Got	
✔	11 10 9 7 6 5 1 2 3 8 4 7	7	7	✔
✔	5 1 2 3 4 4	4	4	✔
✔	5 1 1 2 3 4	1	1	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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